

Core Logic Design of Trading Strategies		
Targeting Market	FIFA World Cup match outcome betting markets. Pre-match win/draw/loss wagers on regulated sportsbooks.	
Phenomena Description ☒	Theoretical Framework ☐	Betting markets for World Cup matches are disproportionately influenced by team reputation, media narratives, and casual bettor sentiment. This leads to systematic mispricings, especially around upsets where a lesser-known team has strong underlying player-level statistics but low public perception.
Analytical Framework ☒	Theoretical Hypothesis ☐	A fine-tuned LLaMA 3.1 8B model (QLoRA) ingests structured player-level statistics from club leagues and international tournaments for both teams. The model outputs match outcome predictions (win/draw/loss, predicted score, match events) along with a natural-language reasoning chain referencing specific player data. This reasoning chain makes every prediction auditable and debuggable, unlike black-box ML approaches.
Phenomena Interpretation ☒	Theoretical Reasoning ☐	World Cup matches attract high volumes of sentiment-driven bettors. The matches happen infrequently (every 4 years), so historical data for national team matchups is thin, and national pride distorts risk assessment. These behavioral factors create persistent gaps between market-implied probabilities and data-driven estimates grounded in player performance across domestic leagues.
Trading Opportunity	When the model's predicted probability for a match outcome diverges significantly from the market's implied probability (or from conventional wisdom), this gap represents exploitable alpha. The model identifies value bets by grounding predictions in granular player data rather than reputation or narrative. Backtested against 2022 FIFA World Cup results (64 matches) as ground truth.	
Trading Operations	Pre-match only. For each World Cup match: (1) generate prediction with reasoning chain from player stats, (2) human reviews reasoning for factual errors such as injuries or suspensions, (3) if predicted edge exceeds the confidence threshold, size position via fractional Kelly criterion, (4) place bet before kickoff. Hard cap on maximum exposure per match. Drawdown circuit breaker pauses activity if cumulative losses exceed a predefined bankroll percentage.	

Financial Instrument(s)	Fixed-odds sports betting contracts (win/draw/loss) on regulated sportsbooks. Potential extensions include Asian handicap lines and over/under (total goals) markets.
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Table 1: Core Logic Design of Trading Strategies